Makeup Exam

Lab Evaluation-I

Data Analytics Using Python (CSE2101)

Total Marks= 20 [CO2]

Date: Sep 17, 2025
Duration: 50 mins

Name:_________________________

Roll No:_________________ **PC No:**________

Section:__________

Q1: Suppose you have an array of 12 monthly sales values (in ₹1000):

sales = [12, 15, 14, 10, 8, 20, 22, 18, 25, 30, 28, 35]

I. Convert the list into a **NumPy array** and reshape it into 3 rows × 4 columns. [2]
II. Find the **mean, median, and standard deviation** of sales. [2]
III. Using **Matplotlib**, plot a **line graph** of monthly sales. [2]
IV. Using **Seaborn**, plot a **boxplot** of sales values. [2]

Q2. Using **Pandas** and **NumPy**, generate a DataFrame of **10 employees** with random values for the following columns:

- **Emp_ID** (unique, 201–210)
- **Name** (randomly pick from: ["Amit", "Riya", "Sohan", "Neha", "Karan", "Pooja", "Manoj", "Isha", "Vikas", "Divya"])
- **Department** (random choice from ["HR", "IT", "Finance", "Sales"])
- **Age** (random integer between 22–45)
- **Salary** (random integer between 35,000–90,000)
- **Experience** (random integer between 1–15 years)

I. Write code to generate the DataFrame with random values. [6]

II. Add a new column Performance_Level such that: [2]

- If Salary > 70,000 and Experience > 10 → "Excellent"
- If Salary > 50,000 → "Good"
- Else → "Average"

III. Display employees whose **Age is above the average Age of the company**. [2]

IV. Identify relation between **Experience** and **Salary**. Briefly interpret the trend (if employees with more experience earn higher salaries). [2]

Note: *Make sure all plots should have proper aesthetics and clearly readable.*